LESSON PLAN

EMI/EMC (14)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	07
Total No. of Lesson Plans	46	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Introduction to circular position lines, visual and radar range position lines and hyperbolic position lines, Introduction to the use of pos
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Introduction to circular position lines, visual and radar range position lines and hyperbolic position lines, Introduction to the use of position lines for ground speed check, ETA check, track check, heading check and homing.
5. Specific Objectives	14.1.1.1 Trainee will be able to explain and define a position on the earth.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Nil specified; default to L & Ds and IB, PDF
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What is the definition of a position on the earth?
(b) Name the three types of position lines covered in this lesson.
(c) Why are position lines essential for navigation?

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Definition of a position on the earth • Circular position lines concept • Visual range position lines • Radar range position lines • Hyperbolic position lines	L & Ds	PPT, IB	15 min
(b)	• Use of position lines for ground speed check • Use of position lines for ETA check • Use of position lines for track check • Use of position lines for heading check • Use of position lines for homing	L & Ds	PPT, PDF	20 min
(c)	• Summary of position line types • Application of checks in flight • Review of specific objectives • Clarification of doubts • Preparation for next lesson	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain how a circular position line is generated.
(b) Describe the difference between visual and radar range position lines.
(c) How can position lines be used to verify ground speed?

12. Home Assignment and Follow Up	Prepare a brief note on how the concepts of this lesson will be applied during the practical demonstration on the Globe in LP 08.
Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Practical demonstration on Globe (Earth's Model) and how to do compass swing.	Refer to the master syllabus reference list	LP 08

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min